# China-related Subreddit Stance Analyzer

ICSS Final Project Report
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## 1. Motivation

The motivation for this project begins with a famous dispute in Chinese-speaking communities—not only in the PRC, but also in other Chinese-speaking regions—or, more precisely, with a widely spread hypothesis: The negative impressions of foreign observers toward “China,” understood as the nation in an abstract cultural or geographic sense rather than as a clearly defined political entity, and toward “Chinese,” which is likewise a loosely defined anthropological term, should be attributed to their negative impressions of the regime of the modern People’s Republic of China. In short, this hypothesis can be summarized as follows: the negative views of outside observers are “anti-communist, not anti-China.” This hypothesis can be broken down into two points: first, foreign observers hold **negative** views toward China and Chinese people; second, that there is a **causal** relationship between attitudes toward China and attitudes toward its regime.

For that reason, this project aims to answer the following research question: when posts in China-related subreddits mention the PRC regime, are there any significant differences in their stances?

The motivation is both analytical and technical. Analytically, this project aims to software to examine the hypothesis discussed above and present the distribution of stances in a visible and interactive way. Technically, I wanted to build a reproducible pipeline that goes beyond this specific topic and can move from raw social-media text to labeled outputs and an interactive UI.

## 2. Design

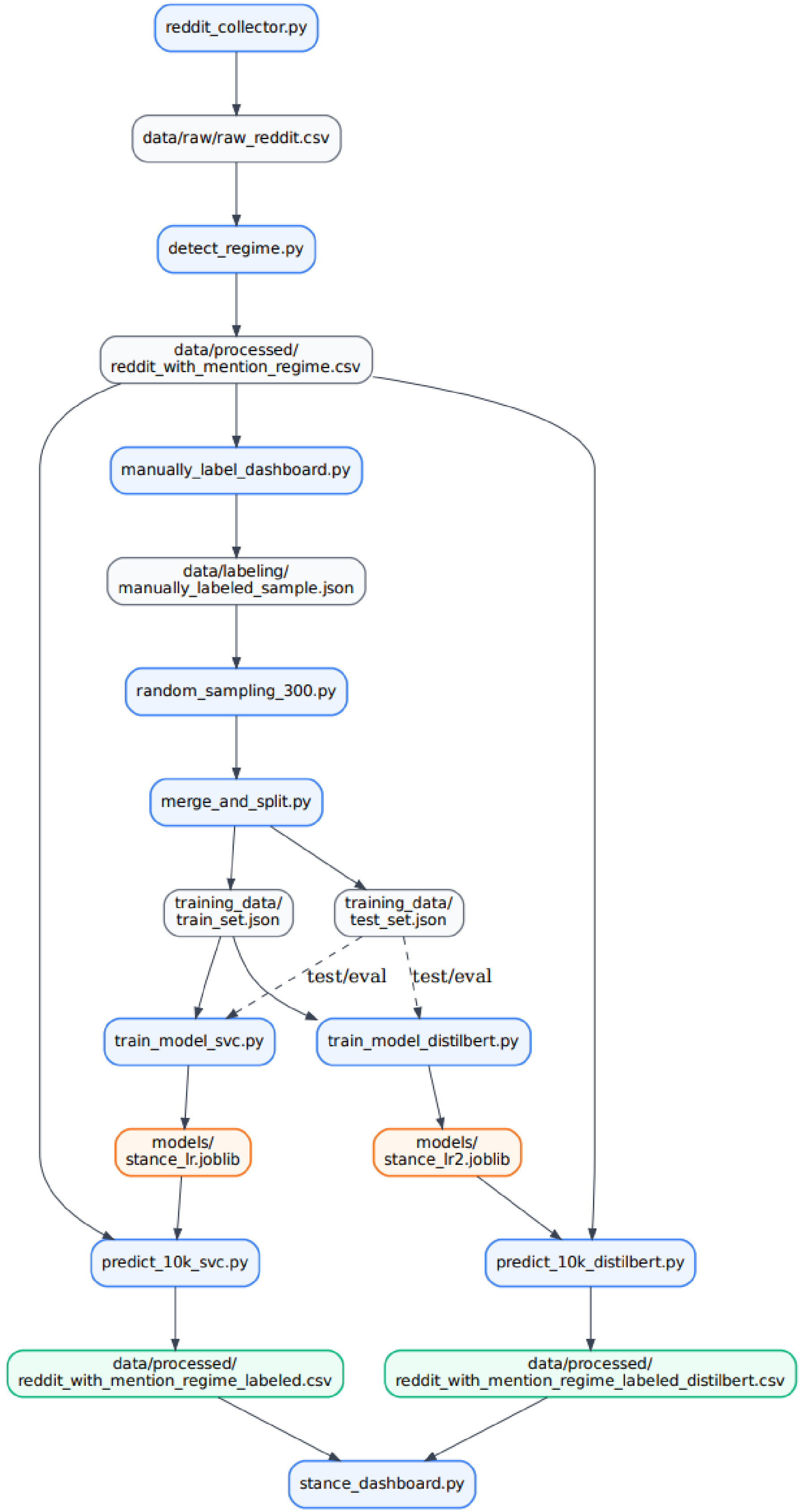
reddit_collector.py
data/raw/raw_reddit.csv
detect_regime.py
data/processed/
reddit_with_mention_regime.csv
manually_label_dashboard.py
data/labeling/
manually_labeled_sample.json
random_sampling_300.py
merge_and_split.py
training_data/
train_set.json
training_data/
test_set.json
test/eval
test/eval
train_model_svc.py
train_model_distilbert.py
models/
stance_lr.joblib
models/
stance_lr2.joblib
predict_10k_svc.py
predict_10k_distilbert.py
data/processed/
reddit_with_mention_regime_labeled.csv
data/processed/
reddit_with_mention_regime_labeled_distilbert.csv
stance_dashboard.py

*Figure 1. Overall workflow of the project pipeline.*

## 2 Main modules

- **reddit_collector.py:** Collects posts from selected subreddits through Reddit's public JSON endpoint and stores a standardized raw CSV.
- **detect_regime.py:** Adds new Boolean variable "regime_mention" by detecting regime related expressions

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    communist\s+party\s+of\s+china|
    party[-\s]?state|
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# core politicians and supreme political centre
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    xi(\s+jinping)?|
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# --- State organs
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# regime-issue
|\b(
    tiananmen|june\s+4(th)?|
    xinjiang|uyghur(s)?|
    tibet(an)?|
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*Figure 2. Regular expression for detecting regime mention*

- **random_sampling_300.py:** extracts 300 posts randomly for labeling.
- **manually_label_dashboard.py:** Provides a Streamlit interface for one-by-one stance annotation with progress saved to disk, which allows me to annotate 300 posts with "critical" "neutral" or

"positive". This part is pasted from the dashboard with the similar functions written by myself for assignment 3.

It is worth mentioning that my labeling logic is that critical, neutral and positive are not considered as a whole, but specifically refer to the attitude towards China and Chines. For example, some posts pointed out that the U.S. human rights agencies' accusations against China were unfair. The overall attitude of such posts seemed to be critical, but because they expressed a protest against the stigmatization of China, they were also labeled as positive (toward China) by me.

- **merge_and_split.py:** The manual labeled sample are saved in manually_labeled_sample.json. From which, the two modules draw a random sample of 300 posts, and prepares the training and test sets.
- **train_model_svc.py / train_model_distilbert.py :** The project then trains two stance classification models: an SVC model through and a DistilBERT-based model through.
- **predict_10k_svc.py / predict_10k_distilbert.py:** Then the project trys to predict and label the majority of the posts with its stances.

**stance_dashboard.py**: provides an interactive dashboard for exploring the results. It allows users to compare stance distributions, and select subreddits of interest and view (or select all) the distribution of instances under different subreddits.

## 3. Innovation

### 3.1 Use of Reddit JSON endpoint

First,I learned how to collect Reddit data without relying on Reddit API, which has been no more available since last year. I used Reddit's public JSON endpoint and implemented own collection logic with requests. This included building a page turner, setting an User-Agent, and responding to rate limits through backoff and retry logic.

### 3.2 Use of re.compile for detecting references of the regime

There I operalized the theoretical concept (basically, "the mention of the regime") from our research question. I translated it into a reproducible detection rule using re.compile with a large regular-expression pattern, which includes not only direct references to governments, state organs and the Party, but also indirect references, e.g. "sensitive" historical events, and politically regime-related issues. In this way, I tried how to transform an abstract narrative framing concept into a measurable binary variable, whiche is named "mention_regime".

### 3.3 Training different models using pipeline

For the classical machine-learning baseline, I learned how to combine TfidfVectorizer and LinearSVC into a Pipeline to create a reusable workflow. I also learned how to address class imbalance (my data is highly umbalanced) through setting parameter like “class_weight="balanced". The use of a pipeline made the model easier to save, reload, and apply consistently to new data.

In addition, I also explored a transformer-based approach with DistilBERT. This required learning several new toolse: especially building PyTorch standardized interface from the training set, since the Transformer Trainer can only accept training set in given format. Although the DistilBERT results were not that satisfying as the SVC, learning this model was still helpful.

### 3.4 Visualizaiton by Streamlit dashboard

Even though building an interactive dashboard with Streamlit and Plotly is nothing new, I have added some additional functions beyond what we have learned before. My dashboard allows users to see the distribution of stances grouped by assigned subreddits. It is really enlightening to see the differences not only between posts that do not mention the regime and those that do, but also between posts from different groups.

## 4. Results

### 4.1 Models

For our models, overall, the SVC version performs better than DistilBERT on the manually labeled test set. The SVC model reached an accuracy of 0.70 and a macro F1 score of 0.53, while DistilBERT achieved the same overall accuracy but a clearly lower macro F1 score of 0.39. This indicates that, under the current data conditions (only around 300 manually labeled posts in total, with a 10% test split), the classical SVC pipeline provided a more balanced classification performance than the transformer-based model. At the same time, both models showed a strong tendency to predict the neutral class, which is not surprising since the data is highly unbalanced as the most posts from these Subreddits should rather be seen as neutral discussion or information sharing, rather than disputes.

### 4.2 conclusion

The visualization shows a clear overall pattern: across subreddits, posts that mention the regime contain a **lower share of neutral labels and higher shares of both critical and positive labels than posts that do not mention the regime.** This suggests that once discussion shifts from “China” in a broad sense to the regime more specifically, the discourse becomes less neutral and more explicitly evaluative.

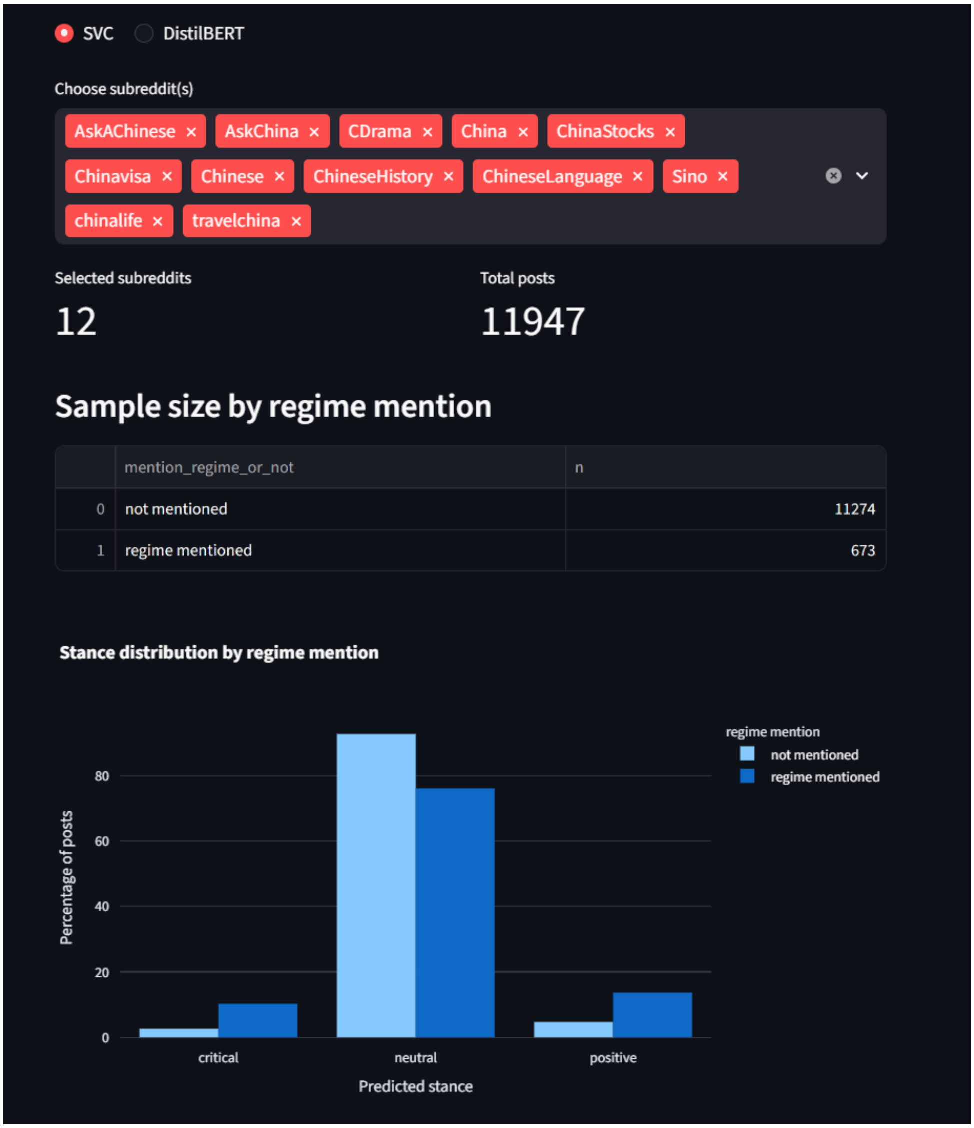


*Figure 3. Distributions of stances from all 12 China related Subreddits*

At the same time, this pattern is not equally strong across subreddit types. In culture- and lifestyle-related subreddits (e.g. CDrama ), the difference between the two groups is relatively small, and most posts remain overwhelmingly neutral regardless of whether the regime is mentioned.

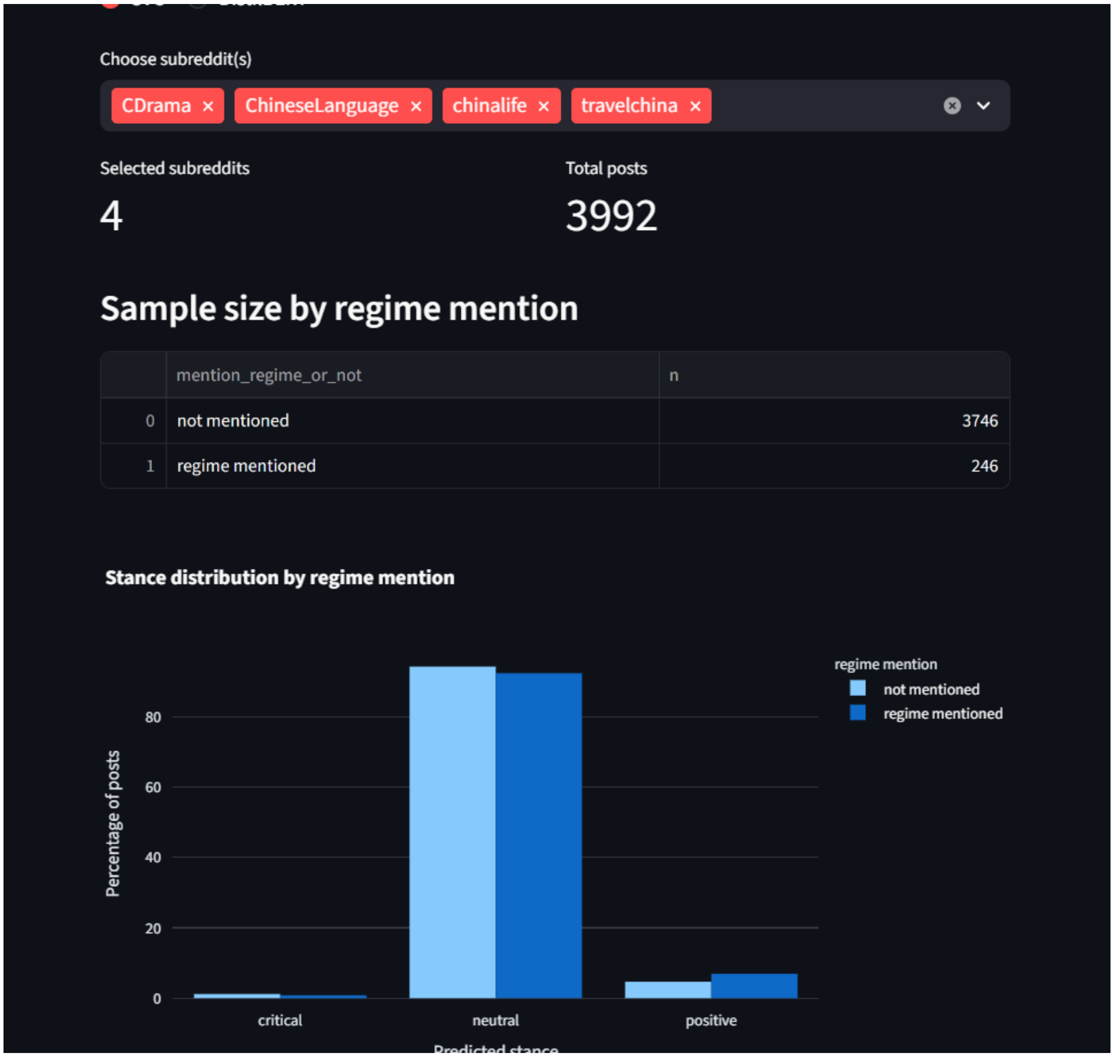


*Figure 4. Distributions of stances from* culture- and lifestyle-related subreddits

By contrast, in the more political, historical, economic, or broadly issue-related subreddits, the difference is much more visible: posts mentioning the regime show a noticeably lower neutral share and higher proportions of critical and positive labels. In other words, regime metioning appears to matter much more in these kinds of discussions than in everyday cultural discussions.

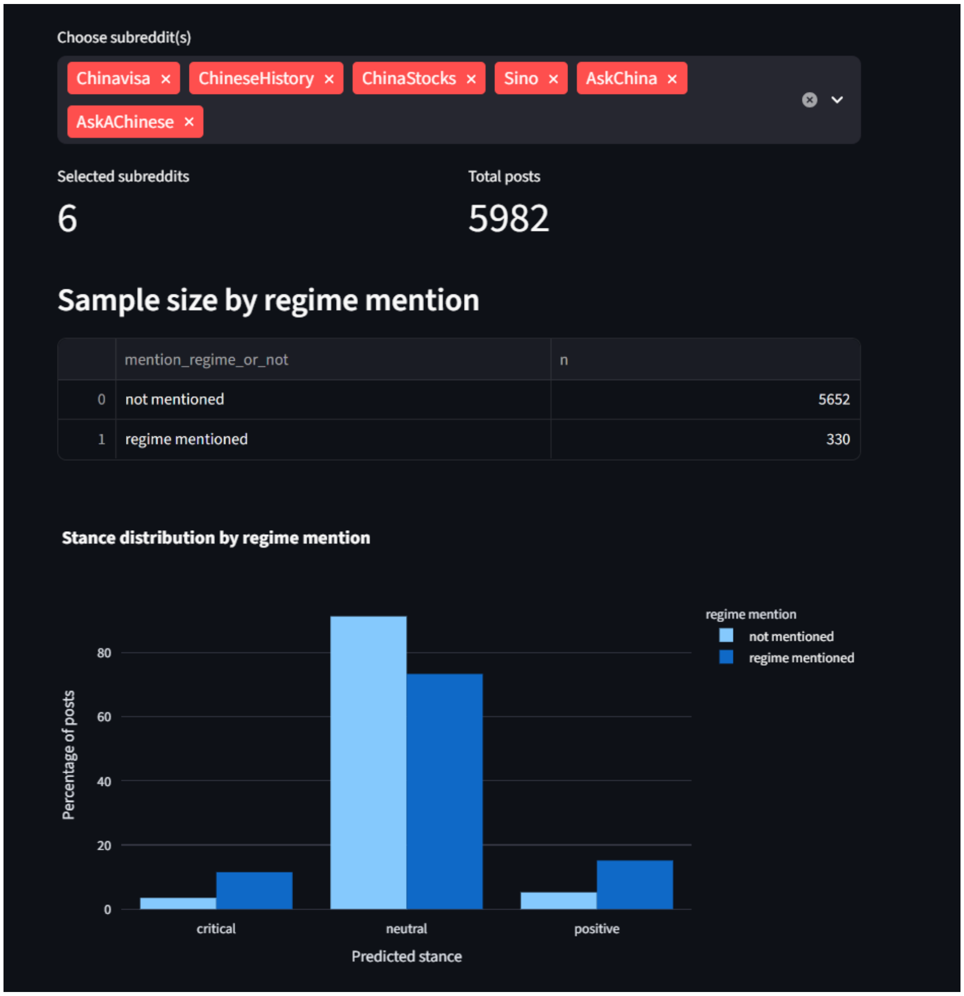


*Figure 5.*

*Distributions of stances from* political, historical, economic, or broadly issue-related subreddits

## 5. Outlook

As for the challenges encountered, I think the biggest problem is preparing the training sets required for training. First of all, due to the inefficiency of manual labeling and annotation, three hundred annotations consume a lot of time and are still too few for training the model. Secondly, I must admit that the judgment criteria I used for manually labeling are not scientific enough, or lack unified and clear judgment criteria. As mentioned above, the basis for manual labeling is the stance of the post toward China and Chinese. But more often than not, many posts do not clearly reflect any obvious attitude towards China, but only comment on Chinese products or TV series - that is to say, those posts that clearly express praise for Chinese society (such as public security and technological development), and those that express love for Chinese idols, will be labeled as positive by me. This may explain why there are more positive posts in subreddits related to life and culture. I intuitively feel that the latter is also a cultural factor that should be generally considered, but there is a lack of relevant academic theoretical support. For improvement, LLMs can be introduced through API (e.g. DeepSeek API) for massively labeling.

Another practical challenge throughout the project was file-path management. Because the workflow involved multiple files across different folders, and development environment always moved between Windows and WSL/Linux, paths often broke and outputs were not always saved or loaded from the intended location. This repeatedly interrupted training and dashboard development, and showed that reproducibility depends not only on the model itself but also on a well organized project structure. In future work, this could be improved by standardizing the directory layout, using more robust relative paths.

Going beyond, the overall workflow and infrastructures of this project is not limited to the topic of China. Although the current case focuses on China-related Reddit discussions, the project is transferable to many other topics that involve narrative framing and stance analysis. In principle, the same structure—data collection from Reddit, keyword-based detection of a specific topic, manual labeling( or maybe automatically label by LLMs), supervised training, and interactive visualization—could be adapted to debates about other countries, political actors, social movements, or controversial public issues. With different keywords, labels, and better training set, the same system could be applied to questions such as attitudes toward immigration, climate policy, etc. Future work could therefore focus on turning the current project into a more general framework for analyzing how different narrative frames shape online attitudes across topics.